

Q.1. Find determinant :-

$$1. \begin{vmatrix} 5 & 7 \\ -3 & -2 \end{vmatrix} = -10 + 21 \\ = 11$$

$$2. \begin{vmatrix} 1 & -2 \\ 3 & 4 \end{vmatrix} = 4 + 6 \\ = 10$$

$$3. \begin{vmatrix} 3 & -1 \\ 2 & 3 \end{vmatrix} = 9 + 2 \\ = 11$$

$$4. \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} = 0 + 1 \\ = 1$$

$$5. \begin{vmatrix} 1 & 8 & 1 \\ 2 & -5 & 2 \\ 3 & 7 & 3 \end{vmatrix} \quad R_1 = 1(-15 - 14) - 8(6 - 6) + 1(14 + 15) \\ = 1(-29) - 8(0) + 1(29) \\ = -29 + 29 \\ = 0$$

$$6. \begin{vmatrix} 0 & 5 & 0 \\ 2 & 6 & 8 \\ 5 & -6 & 2 \end{vmatrix} \quad R_1 = 0(12 + 48) - 5(4 - 40) + 0(-12 - 30) \\ = 0 - 5(-36) + 0 \\ = 180$$

$$7. \begin{vmatrix} 5 & -6 & 8 \\ 2 & -3 & 4 \\ 7 & -2 & 4 \end{vmatrix} \quad R_1 = 5(-12 + 8) + 6(8 - 28) + 8(-4 + 21) \\ = 5(-4) + 6(-20) + 8(17) \\ = -20 - 120 + 36 \\ = -140 + 136 \\ = -4$$

$$8. \begin{vmatrix} -1 & 6 & -2 \\ 5 & 0 & 7 \\ 4 & 1 & -3 \end{vmatrix} \quad R_1 = -1(0 - 7) - 6(-15 - 28) - 2(5 - 0) \\ = -1(-7) - 6(-43) - 2(5) \\ = 7 + 258 - 10 \\ = 265 - 10 \\ = 255$$

Q.2 Find co-factor :-

1.	$\begin{vmatrix} 5 & 3 & 1 \\ 4 & -3 & 0 \\ 6 & 1 & 2 \end{vmatrix}$	$M_{11} = -6$	$A_{11} = -6$
		$M_{12} = 8$	$A_{12} = -8$
		$M_{13} = 22$	$A_{13} = 22$
		$M_{21} = 5$	$A_{21} = -5$
		$M_{22} = 4$	$A_{22} = -4$
		$M_{23} = -13$	$A_{23} = 13$
		$M_{31} = 3$	$A_{31} = 3$
		$M_{32} = -4$	$A_{32} = 4$
		$M_{33} = 27$	$A_{33} = 27$

2.	$\begin{vmatrix} 1 & 7 & -3 \\ -4 & 6 & 2 \\ 2 & -5 & 3 \end{vmatrix}$	$M_{11} = 28$	$A_{11} = 28$
		$M_{12} = -16$	$A_{12} = 16$
		$M_{13} = 8$	$A_{13} = 8$
		$M_{21} = 6$	$A_{21} = -6$
		$M_{22} = 9$	$A_{22} = 9$
		$M_{23} = -19$	$A_{23} = 19$
		$M_{31} = 32$	$A_{31} = 32$
		$M_{32} = -10$	$A_{32} = 10$
		$M_{33} = 34$	$A_{33} = 34$

3.	$\begin{vmatrix} 5 & -6 & 8 \\ 2 & -3 & 4 \\ 7 & -2 & 4 \end{vmatrix}$	$M_{11} = -4$	$A_{11} = -4$
		$M_{12} = -20$	$A_{12} = 20$
		$M_{13} = 17$	$A_{13} = 17$
		$M_{21} = -8$	$A_{21} = 8$
		$M_{22} = -36$	$A_{22} = -36$
		$M_{23} = 32$	$A_{23} = -32$
		$M_{31} = 0$	$A_{31} = 0$
		$M_{32} = 4$	$A_{32} = -4$
		$M_{33} = -3$	$A_{33} = -3$

4. $\begin{vmatrix} -1 & 6 & -2 \\ 5 & 0 & 7 \\ 4 & 1 & -3 \end{vmatrix}$

$M_{11} = -7$	$A_{11} = -7$
$M_{12} = -43$	$A_{12} = 43$
$M_{13} = 5$	$A_{13} = 5$
$M_{21} = -16$	$A_{21} = 16$
$M_{22} = 11$	$A_{22} = 11$
$M_{23} = -25$	$A_{23} = 25$
$M_{31} = 42$	$A_{31} = 42$
$M_{32} = 3$	$A_{32} = -3$
$M_{33} = -30$	$A_{33} = -30$

5. $\begin{vmatrix} 0 & 5 & 0 \\ 2 & 6 & 8 \\ 5 & -6 & 2 \end{vmatrix}$

$M_{11} = 60$	$A_{11} = 60$
$M_{12} = -36$	$A_{12} = 36$
$M_{13} = -42$	$A_{13} = -42$
$M_{21} = 10$	$A_{21} = -10$
$M_{22} = 0$	$A_{22} = 0$
$M_{23} = -25$	$A_{23} = 25$
$M_{31} = 40$	$A_{31} = 40$
$M_{32} = 0$	$A_{32} = 0$
$M_{33} = -10$	$A_{33} = -10$

6. $\begin{vmatrix} 1 & 8 & 1 \\ 2 & -5 & 2 \\ 3 & 7 & 3 \end{vmatrix}$

$M_{11} = -29$	$A_{11} = -29$
$M_{12} = 0$	$A_{12} = 0$
$M_{13} = 29$	$A_{13} = 29$
$M_{21} = 17$	$A_{21} = -17$
$M_{22} = 0$	$A_{22} = 0$
$M_{23} = -17$	$A_{23} = 17$
$M_{31} = 21$	$A_{31} = 21$
$M_{32} = 0$	$A_{32} = 0$
$M_{33} = -21$	$A_{33} = -21$

$$\begin{array}{r} 2 \times 2 \\ 4 \times 1 \\ \hline 4 \times 1 = 5 \end{array}$$

11 Q₃

$$\boxed{11} \quad \begin{vmatrix} 6 & -2 \\ 9 & x \end{vmatrix} = 0$$

$$\Rightarrow 6x + 18 = 0$$

$$\Rightarrow 6x = -18$$

$$x = \frac{-18}{6}$$

$$\boxed{x = -3}$$

$$\boxed{12} \quad \begin{vmatrix} x-3 & 1 \\ 2 & x-2 \end{vmatrix} = 0$$

$$\therefore (x-3)(x-2) - 2 = 0$$

$$\therefore x^2 - 2x - 3x + 6 - 2 = 0$$

$$\therefore x^2 - 5x + 4 = 0$$

$$\therefore x^2 - 4x - x + 4 = 0$$

$$\therefore x(x-4) - 1(x-4) = 0$$

$$\therefore (x-1)(x-4) = 0$$

$$\therefore x-1 = 0 \quad \text{and} \quad x-4 = 0$$

$$\therefore \boxed{x = 1} \quad \text{and} \quad \boxed{x = 4}$$

$$\boxed{13} \quad \begin{vmatrix} 5 & x \\ 4 & 3 \end{vmatrix} = -1$$

$$\Rightarrow 15 - 4x = -1$$

$$\Rightarrow -4x = -1 - 15$$

$$\Rightarrow -4x = -16$$

$$\Rightarrow 4x = 16$$

$$\Rightarrow x = \frac{16}{4}$$

$$\Rightarrow \boxed{x = 4}$$

$$\boxed{14} \quad \begin{vmatrix} x+3 & 2 \\ 3 & x-3 \end{vmatrix} = 1$$

$$\Rightarrow 0 \therefore x^2 - 9 - 6 = 1$$

$$\therefore x^2 - 15 = 1$$

$$\therefore x^2 = 1 + 15$$

$$\therefore x^2 = 16$$

$$\therefore x = \pm 4$$

$$\therefore \boxed{x = 4} \text{ and } \boxed{x = -4}$$

$$\boxed{15} \quad \begin{vmatrix} 4 & 1 & -2 \\ 4 & 4 & +2 \\ 1 & 3 & 1 \end{vmatrix} = 0$$

$$\therefore 4(4 - 6) - 1(4 - 2) - 2(12 - 4) = 0$$

$$\therefore 4(-2) - 1(2) - 2(8) = 0$$

$$\therefore -8 - 2 - 16 = 0$$

$$(1) \therefore -24 - 18 = 0$$

$$\therefore -24 = 18$$

$$\therefore -41 = 18$$

$$\therefore -24 + 18 = -6$$

$$\therefore \boxed{-4 = -9}$$

$$\boxed{16} \quad \begin{vmatrix} x & 2 & 3 \\ 5 & 0 & 7 \\ 3 & 1 & 2 \end{vmatrix} = 30$$

$$\therefore x(0 - 7) - 2(10 - 21) + 3(5) = 30$$

$$\therefore -7x - 2(-11) + 15 = 30$$

$$\therefore -7x + 22 + 15 = 30$$

$$\therefore -7x = 30 - 15 - 22$$

$$\therefore -7x = -7$$

$$x = -7/-7$$

$$\boxed{x = 1}$$

$$\boxed{17} \quad \begin{vmatrix} x-2 & 2 & 2 \\ -1 & x & -2 \\ 2 & 0 & 4 \end{vmatrix} = 0$$

$$\therefore (x-2)(4x) - 2(-4+4) + 2(0-2x)$$

$$\therefore 4x^2 - 8x - 4x = 0$$

$$\therefore 4x^2 - 12x = 0$$

$$\therefore 4x(x-3) = 0$$

$$\therefore 4x = 0 \quad \text{and} \quad x-3 = 0$$

$$\therefore \boxed{x=0} \quad \text{and} \quad \boxed{x=3}$$

$$\boxed{18} \quad \begin{vmatrix} x-2 & 2 & 1 \\ x & 1 & x+1 \\ 1 & 1 & 0 \end{vmatrix} = 4$$

$$(x-2)(0 - (x+1)) - 2(0 - (x+1))$$

$$+ 1(x-1) = 4$$

$$(x-2)(-x-1) - 2(-x-1) + x-1 = 4$$

$$-x^2 - x + 2x + 2 + 2x + 2 + x - 1 - 4 = 0$$

$$-x^2 - x + 2x + 2x + x + 2 + 2 - 1 - 4 = 0$$

$$-x^2 + 4x - 1 = 0$$

$$x^2 - 4x + 1 = 0$$

$$a = 1, \quad b = -4, \quad c = 1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{4 \pm \sqrt{16 - 4}}{2}$$

$$= \frac{4 \pm \sqrt{4 \times 3}}{2} \Rightarrow \frac{4 \pm 2\sqrt{3}}{2}$$

$$x = \frac{4 + 2\sqrt{3}}{2} \quad \text{and} \quad \frac{4 - 2\sqrt{3}}{2}$$

$$x = 2 + \sqrt{3} \quad \text{and} \quad \cancel{x = 2 - \sqrt{3}}$$

9	x	4	4	= 0
	4	x	4	
	4	4	0	

$$\therefore 0(0 - 16) - 4(0 - 16) + 4(16 - 4x) = 0$$

$$\therefore -16x + 64 + 64 - 16x = 0$$

$$\therefore -32x + 128 = 0$$

$$\therefore -32x = -128$$

$$\therefore 32x = 128$$

$$\therefore x = \frac{128}{32}$$

$$\therefore x = 4$$

10	3	4	8	= 0
	x-3	-8	-16	
	3	9	4	

$$\therefore 3(-32 + 144) - 4(4x - 12 + 48) + 8(9x - 27 + 24) = 0$$

$$\therefore 3(112) - 4(4x + 36) + 8(9x - 3) = 0$$

$$\therefore 336 - 16x - 144 + 72x - 24 = 0$$

$$\therefore 56x + 392 = 0$$

$$\therefore 56x = -392$$

$$\therefore x = \frac{-392}{56}$$

$$x = -7$$